# Aaryan S. Maralihalli

Healthcare / Biosignals Machine Learning

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## Summary

**IEEE-published**

*ML engineer for healthcare/biosignals — built an interpretable **ECG/PPG** model for stroke prediction & arrhythmia classification (**MIMIC-III, ROC-AUC 0.757**) with clinician-facing explainability. Fluent with HIPAA-grade medical data. CS graduate, Jul 2026.*

## Publication

- **Aaryan S. Maralihalli et al.**, *"An Interpretable Model for Stroke Prediction and Arrhythmia Classification using ECG & PPG,"* **IEEE ICASI 2026**, Kyoto — **Oral Presentation** (Paper J260481).

## Healthcare ML Work

- **Self-Supervised Cardiac Risk System (capstone / published): contrastive learning** on **MIMIC-III** ECG/PPG (133K segments, 8 channels, 288 patients); 1D-CNN **WaveformEncoder** + Random Forest clinical model over 14 features; **ROC-AUC 0.757**, recall 0.57 on the rare (~5%) stroke class; benchmarked against the **CHA2DS2-VASc** clinical score.
- **Explainability for clinicians:** temporal saliency maps, channel importance, and integrated gradients; **FastAPI** inference + Streamlit dashboard generating clinician-facing risk reports.
- Arrhythmia classification (N/S/V/F/U) with SHAP-based interpretability.

## Experience

**Integrated Genetic Solutions (IGS), Bangalore**
*Project Intern, Medical AI* Jun – Aug 2024

- Worked on **HIPAA-compliant medical-AI** workflows extracting lab values from blood reports across major labs (Quest, LabCorp, BioReference); supported the dashboard migration handling **5M+ lab values**.

**HyperVerge, Bangalore**
*Integration Engineer Intern* Jan 2026 – Jun 2026

- Applied production AI/ML models (face match, liveness) for regulated fintech clients.

## Skills

- **Biosignals/Clinical:** ECG/PPG signal processing, MIMIC-III, arrhythmia & stroke modeling, clinical validation (CHA2DS2-VASc), HIPAA-aware handling
- **ML/DL:** PyTorch, contrastive/self-supervised learning, CNNs, Random Forest, XAI (saliency, integrated gradients, SHAP), scikit-learn
- **Engineering:** Python, FastAPI, NumPy/Pandas, HDF5, Docker

## Education

- **B.Tech, Computer Science** — PES University, Bangalore — 2022 – Jul 2026
- *Certifications:* CITI Program (Conflicts of Interest; Data/Specimens Only Research)